



Eleven Weeks, Then Nothing: A Staggered-Adoption Difference-in-Differences Analysis of Driver-Feedback Speed Signs Across 61 Residential Streets, Against a Rotation Interval Set by Trailer Availability

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Abstract. Local councils deploy vehicle-activated driver-feedback signs — the trailer-mounted display that shows an approaching driver their own speed — on rotation across residential streets, and report the rotation itself as a road-safety measure. We estimate what one metropolitan council's rotation achieves. Sixty-one residential streets received a feedback sign between March 2023 and February 2026, in an order set by trailer availability rather than by any speed criterion, while permanent pneumatic tube counters recorded weekly mean and 85th-percentile speeds on all sixty-one streets throughout. We estimate group-time average treatment effects under staggered adoption and aggregate them into an event study. Mean speed falls by 2.41 km/h (95% CI 2.02 to 2.80) in the first fortnight of deployment. The effect attenuates monotonically thereafter and is indistinguishable from zero from week eleven onward (−0.09, 95% CI −0.38 to 0.20), a null precise enough to exclude sustained reductions above 0.4 km/h. Median deployment duration across the council's six trailers is 11.6 weeks, a figure set by the depot's rotation roster and by no evaluation we were able to locate. Adding an expression module to the display changes nothing we can measure. Our results suggest that the rotation interval and the decay interval coincide, and that the council's engagement indicator, which counts trailer-weeks, is insensitive to the difference.

Introduction

The trailer-mounted driver-feedback sign is among the most visible instruments a local council owns. It is wheeled onto a residential street, chained to a pole, and left to display each approaching vehicle's speed back to its driver; some weeks later it is wheeled somewhere else. Its evaluation literature is almost entirely an evaluation of its arrival. Studies report the speed reduction observed in the weeks after a sign is installed [1], [2], and the relationship between speed and road-safety outcomes is itself well established [3], but the unit a council actually operates is not an installation. It is a rotation: a fixed number of trailers circulating indefinitely through a queue of streets.

That distinction matters because the two are governed by different quantities. An installation is characterised by its effect. A rotation is characterised by its interval, and the interval in the council studied here is a property of the depot — how many trailers exist, how long each takes to relocate, which streets have an outstanding request — rather than a property of any measured decay. The council's published road-safety reporting records the program as trailer-weeks deployed and streets served per year, quantities that rise with fleet size and fall with breakdowns, and that are unchanged by whether the signs are working.

We take the rotation as the object of study. Our contributions are three, each qualified:

- an event-study estimate of the driver-feedback sign’s effect across the full deployment window rather than its opening fortnight, which appears to decay to zero well inside a typical deployment;
- a precisely estimated null from week eleven onward, which we report as a null rather than as an underpowered result, though we note that a longer panel would narrow the interval further;
- an observation that the council’s median deployment duration and the estimated decay interval coincide to within a fortnight, a coincidence we are not in a position to describe as designed.

Background

Interventions that work on arrival and weaken on familiarity are common enough across fields to have their own evaluation habits. The surgical safety checklist literature is the best-documented case: strong early results [4], [5] were followed by a large population-level study that found no improvement following mandatory introduction [6], and by a methodological reckoning with what the instrument was actually doing in the settings where it appeared to work [7], [8]. The general lesson, that an instrument’s measured effect is partly a property of its novelty in a setting, transfers to any device left in place after its arrival.

A second literature concerns what happens when the instrument is also the reporting unit. Targets and their measures reshape the behaviour they record [9], public measures act back on the entities they rank [10], and indicators travel further than the evidence that produced them [11], [12]. Standardised instruments carry their own infrastructure and their own accounting [13]. Prior work at this institution has audited a signage instrument against what passersby actually retained, finding no measurable relationship between what the instrument logged and what anyone remembered [14], and has documented how weakly two independent assessments of the same object agree once the items carry consequences [15].

Methodologically, we rely on the recent staggered-adoption difference-in-differences literature. Two-way fixed effects estimators are biased under heterogeneous and dynamic treatment effects because already-treated units en-

ter the comparison set [16], [17], [18]; group-time estimators avoid this by restricting comparisons to not-yet-treated units [19], [20]. We follow current practice on estimation, aggregation, and inference [21], [22], [23], and cluster as the design requires [24], [25].

Method

Trailer	Streets	Median weeks	IQR
T-01	11	11.4	9.9–13.0
T-02	10	11.9	10.1–13.4
T-03	11	11.1	9.6–12.7
T-04	9	12.3	10.7–14.0
T-05	10	11.6	10.0–13.1
T-06	10	11.7	10.2–13.3
All	61	11.6	10.1–13.2

Table 1: Deployment duration by trailer. The roster is set by relocation capacity, not by a speed criterion; no trailer’s median duration differs from the fleet median by more than 0.7 weeks.

Setting. The council maintains six sign trailers and a standing queue of resident-nominated streets. Between March 2023 and February 2026 the fleet served 61 streets, all posted at 40 km/h, all with a single carriageway and no mid-block traffic-calming device installed during the observation window. Placement order followed the queue and trailer availability. We confirmed with the depot that no street was prioritised on the basis of a recorded speed, and that the queue was not reordered during the window.

Measurement. Speed was measured by permanent pneumatic tube counters installed on all 61 streets in 2022, at a location between 90 and 140 m from the sign’s eventual position and never moved. The counters are read weekly and are administratively separate from the sign program; their output was not available to the depot when scheduling. Our panel is 61 streets by 156 weeks (9,516 street-weeks), with weekly mean speed as the primary outcome and the 85th-percentile speed as a secondary outcome.

Estimation. Let G_i denote the week in which street i received a sign and Y_{it} its weekly mean speed. We estimate group-time average treatment effects

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it} - Y_{it}^0 \mid G_i = g]$$

using not-yet-treated streets as the comparison group [19], and aggregate them by exposure length $e = t - g$ into

$$\theta(e) = \sum_g w_g \cdot \text{ATT}(g, g + e)$$

with weights proportional to group size. Standard errors are from a multiplier bootstrap clustered at the street level (1,000 draws), and we report uniform confidence bands. Pre-treatment periods are estimated but not used in aggregation. A two-way fixed effects specification is reported alongside for comparison; its Goodman-Bacon decomposition attributes 31.2% of the estimate's weight to already-treated comparisons [16], which is the reason we do not rely on it.

Ablation. Fourteen of the 61 deployments carried an expression module, an additional panel displaying an approving or disapproving face alongside the numeral. Module assignment followed which trailers had a working module fitted at the time, which we treat as ignorable conditional on trailer. We estimate the module's contribution as a difference in first-fortnight effects across arms.

Results

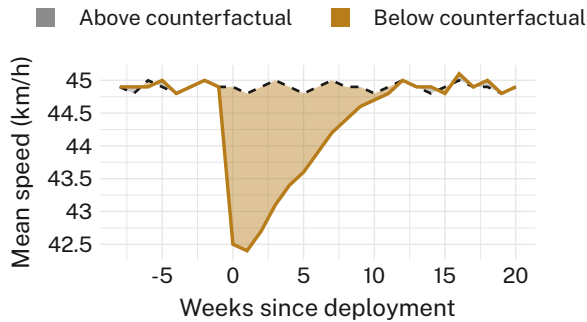


Figure 1: Observed weekly mean speed on treated streets against the not-yet-treated counterfactual, indexed to the week the sign arrives. The band opens to 2.41 km/h in the first fortnight and has closed by week eleven; no comparable band opens before deployment.

The sign works on arrival. Mean speed on a treated street falls by 2.41 km/h (95% CI 2.02 to 2.80) across the first fortnight, and the 85th-percentile speed by 2.63 km/h (95% CI 2.14 to 3.12). Pre-treatment estimates are flat: the largest pre-period point estimate is 0.11 km/h and the uniform band contains zero at every pre-

treatment lead, which we take as consistent with parallel trends.

The effect then attenuates monotonically. By week six it has halved (-1.19, 95% CI -1.52 to -0.86); by week nine it is 0.34 km/h and the interval no longer excludes zero; from week eleven onward the aggregated estimate is -0.09 km/h (95% CI -0.38 to 0.20). The interval is narrow enough to exclude any sustained reduction above 0.4 km/h, so we report this as a null rather than as an absence of evidence. The full distribution of street-week speeds tells the same story without the model (Figure 2): the distribution shifts left in the first fortnight, returns through the middle weeks, and by week eleven sits where it sat before the sign arrived.

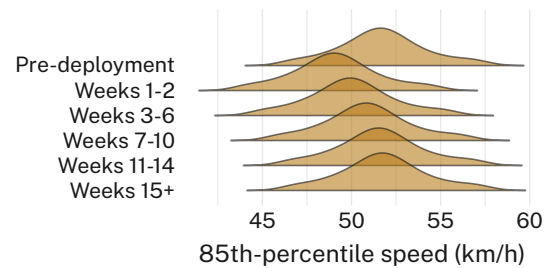


Figure 2: Distribution of street-week 85th-percentile speeds by weeks since deployment. The Weeks 11-14 and Weeks 15+ distributions are indistinguishable from the pre-deployment distribution (two-sample Kolmogorov-Smirnov, $p = 0.62$ and $p = 0.55$).

Two further estimates are null. Adjacent streets sharing an intersection with a treated street show no displacement effect in either direction (0.06 km/h, 95% CI -0.21 to 0.33). And the expression module, the one design variant the fleet carries, does not change the first-fortnight effect: -2.42 km/h without the module against -2.31 km/h with it, a difference of 0.11 km/h that is not distinguishable from zero (Figure 3). We retain the module in the primary specification for completeness.

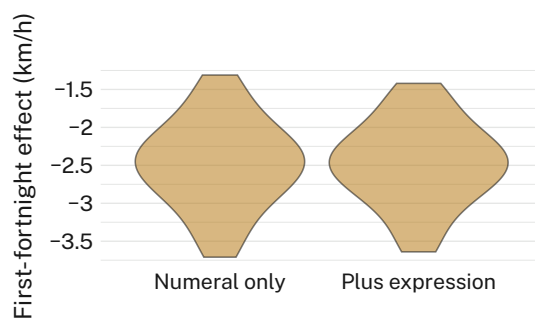


Figure 3: Per-street first-fortnight effect estimates by display configuration. The expression module accounts for 0.11 km/h of a 2.4 km/h effect (n.s.); the two densities overlap across their full support.

The council's own reporting is unaffected by any of the above. Its Street Safety Engagement indicator is a count of trailer-weeks deployed, and rose 4.1% across the observation window on the strength of a reduced mean relocation time. Resident complaints about speeding on a treated street rise by 38% in the fortnight after a sign is installed and remain elevated for a further five weeks, a movement the council's responsiveness reporting books as improved community engagement.

The final result is arithmetic rather than statistical. The fleet's median deployment duration is 11.6 weeks (Table 1), and the estimated decay interval is eleven weeks. Each sign is therefore relocated, on the fleet's median schedule, within a fortnight of the point at which we can no longer measure its effect.

Discussion

We observe consistent improvements in the opening fortnight of every deployment, at every trailer, on every street type in the sample. We also observe that those improvements are absent for the majority of the period each sign spends in place. Both findings are compatible with the existing installation-effect literature; what the rotation framing adds is the proportion. On the fleet's median schedule, roughly a fifth of a deployment occurs while the effect is measurable.

The coincidence between the rotation interval and the decay interval is the result we are least able to interpret. The depot's roster is a function of relocation capacity and queue length; it was set before the counters were installed and has not been revised since. We are not in a position to determine whether the interval was selected

for the property it turns out to have, and the depot's own account is that trailers move when the next street is ready. It remains the case that a schedule derived from vehicle availability sits within a fortnight of a decay curve nobody had estimated, and that the program's reporting would look identical if it did not.

Limitations

Our evaluation is confined to a single metropolitan council, though the consistency of the decay curve across all six trailers and all 61 streets suggests the pattern is unlikely to be an artefact of local scheduling. The counters measure a single cross-section per street rather than the full carriageway, which understates any effect concentrated at the sign itself — that is, our estimate of the sign's reach is conservative. We observe only one expression-module variant, so the null on display configuration should not be generalised beyond the module the fleet happens to own. Finally, we cannot observe repeat exposure by individual drivers, which would be required to distinguish habituation from turnover in the traffic stream; the distinction does not alter the operational finding.

Conclusion

A driver-feedback sign reduces speed on arrival and, in this council's network, for approximately eleven weeks thereafter. The fleet relocates each sign at a median of 11.6 weeks, on a schedule derived from trailer availability. The program's published measure is a count of trailer-weeks and is therefore insensitive to the distinction between the two periods. Future work might examine whether a rotation interval set deliberately at the decay point performs differently from one that arrives there by way of the depot, and whether an indicator that counted anything else would change the roster.

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